

Statistically Validated Invariant-Preserving Physics-Informed Neural Networks for Conservative Dynamical Systems

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Abstract

We present a statistically validated invariant-preserving physics-informed neural network (IP-PINN) framework for long-time integration of conservative dynamical systems, with application to the Lotka–Volterra prey–predator model [1]. Through rigorously controlled experimental protocols involving fifteen independent runs per configuration with fixed random seeding, we demonstrate that embedding analytical invariants within the loss functional yields statistically significant enhancement in solution accuracy relative to baseline physics-informed neural networks (Welch’s one-tailed t-test, $p < 0.001$, Cohen’s $d = 2.59$), while preserving qualitative phase-space structure over long integration times. Our methodology achieves substantial reduction in invariant drift (mean 5.25 vs. 20.7) while incurring negligible computational overhead. Statistical robustness is confirmed through bootstrap confidence interval analysis ($B = 10,000$ replicates), providing a reproducible reference implementation for physics-informed machine learning in dynamical systems. The entire experimental pipeline is publicly archived to ensure complete reproducibility.

Keywords: Physics-informed neural networks, Invariant preservation, Lotka–Volterra systems, Statistical validation, Reproducible research, Long-time integration **MSC (2020):** 65M70; 65P10; 37M05; 68T07

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1. Introduction

Conservative dynamical systems are defined by invariants that govern long-time qualitative behavior. While geometric numerical integration exploits this structure rigorously [2], physics-informed neural networks (PINNs) [3] typically ignore such constraints, leading to uncontrolled invariant drift that corrupts physical fidelity [4, 5, 6, 7, 8].

Prior attempts to address this fall into two categories:

1. Architectural Modification: Hamiltonian neural networks [9] and symplectic networks [10] hard-code geometric structure, limiting flexibility to systems with canonical Hamiltonian form.

2. Heuristic Regularization: Conservative PINNs [11] add invariant penalties without *(i)* theoretical guarantees, *(ii)* statistical validation, or *(iii)* scalability analysis.

We propose a **soft-constrained invariant-penalty framework** that:

- Provides *quantitative* error bounds relating penalty weight to invariant deviation (Section 7);
- Delivers *statistically guaranteed* long-time accuracy through rigorous experimental protocols;
- Scales *without architectural modification* to higher-dimensional systems (Section 8.6).

To our knowledge, this is the first PINN framework that jointly combines invariant penalties, quantitative bounds, and statistically validated long-time performance, establishing a reproducible methodology that bridges classical geometric integration with modern deep learning.

2. Related Work

The PINN literature has expanded rapidly since its inception [3, 7, 8]. Recent advances include adaptive collocation strategies [12], multi-scale architectures [13], accelerated training methods [14], and domain decomposition [15].

Several studies have identified fundamental limitations in long-time integration [4, 5], where residual minimization alone fails to preserve geometric structure. Structure-preserving neural networks have emerged as a remedy,

with conservative PINNs incorporating energy constraints [15, 11], though often requiring Hamiltonian reformulation.

Our IP-PINN differs fundamentally by employing ****soft constraints without architectural modification****, providing statistical guarantees absent in prior work. Unlike conservative PINNs that primarily target energy conservation in Hamiltonian form, our approach applies to general invariants without reformulation.

3. Prey–Predator Model and Invariant

We consider the Lotka–Volterra system:

$$\dot{x} = \alpha x - \beta xy, \quad \dot{y} = \delta xy - \gamma y, \quad (1)$$

where $x(t)$ and $y(t)$ denote prey and predator populations, with parameters $\alpha, \beta, \gamma, \delta > 0$.

System (1) admits the conserved quantity:

$$\mathcal{H}(x, y) = \delta x - \gamma \ln x + \beta y - \alpha \ln y, \quad (2)$$

which remains constant along exact trajectories. Numerical violation of (2) induces artificial damping or growth, corrupting long-term qualitative behavior.

4. Baseline Physics-Informed Neural Network

The baseline PINN approximates solutions $\hat{x}(t; \theta)$, $\hat{y}(t; \theta)$ via a neural network, minimizing:

$$\mathcal{L}_{\text{PINN}} = \mathcal{L}_{\text{ODE}} + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}}, \quad (3)$$

where $\mathcal{L}_{\text{ODE}}$ penalizes ODE residuals and $\mathcal{L}_{\text{IC}}$ enforces initial conditions.

Critically, this formulation omits invariant constraints, permitting solutions that deviate arbitrarily from the invariant manifold $\mathcal{M} = \{(x, y) : \mathcal{H}(x, y) = \mathcal{H}(x_0, y_0)\}$.

5. Invariant-Preserving PINN Method

The IP-PINN augments the loss with an invariant penalty:

$$\mathcal{L}_{\text{INV}} = \mathbb{E}_{t \in [0, T]} [(\mathcal{H}(\hat{x}(t), \hat{y}(t)) - \mathcal{H}(x_0, y_0))^2]. \quad (4)$$

The total loss becomes:

$$\mathcal{L} = \mathcal{L}_{\text{ODE}} + \lambda_{\text{IC}}\mathcal{L}_{\text{IC}} + \lambda_H\mathcal{L}_{\text{INV}}. \quad (5)$$

This penalty formulation continuously enforces proximity to $\mathcal{M}$ during training, analogous to constrained variational problems [16].

6. Statistical Validation Protocol

Following reproducibility guidelines [17, 18], we perform $n = 15$ independent runs per configuration with deterministic seeding ($\text{seed} = 42 + \text{run_id}$). Statistical significance is assessed via one-tailed Welch’s t-test ($\alpha = 0.05$) with variance heterogeneity correction, confirmed by Mann-Whitney U test and percentile bootstrap confidence intervals ($B = 10,000$ replicates). Effect sizes are quantified using Cohen’s d [19] to evaluate practical significance beyond p -values. This protocol provides $> 80\%$ power to detect medium effects ($d \geq 0.5$) while maintaining computational feasibility.

All experimental data, including per-run errors and convergence metrics, are archived in JSON format with complete metadata (see Section 10).

7. Theoretical Justification

We establish quantitative relationships between penalty weight λ_H and invariant violation.

7.1. Invariant-Error Bound

Proposition (Penalty-Drift Relationship). Let $\hat{\mathbf{z}}(t; \theta)$ minimize $\mathcal{L} = \mathcal{L}_{\text{ODE}} + \lambda_{\text{IC}}\mathcal{L}_{\text{IC}} + \lambda_H\mathcal{L}_{\text{INV}}$. Under Lipschitz continuity of $\nabla_{\theta}\mathcal{L}_{\text{ODE}}$ with constant L , the invariant drift satisfies:

$$\mathcal{L}_{\text{INV}}(\hat{\mathbf{z}}) \leq \frac{C}{\lambda_H}, \quad C = \frac{\|\nabla_{\theta}\mathcal{L}_{\text{ODE}}(\theta^*)\|}{L} \quad (6)$$

where θ^* is a stationary point of the unconstrained problem. This bound characterizes training-time behavior rather than post-training numerical integration error.

Proof Sketch.. From optimality $\nabla_{\theta}\mathcal{L} = 0$:

$$\lambda_H \nabla_{\theta} \mathcal{L}_{\text{INV}} = -\nabla_{\theta}(\mathcal{L}_{\text{ODE}} + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}}). \quad (7)$$

By Lipschitz continuity and first-order optimality of θ^* :

$$\|\nabla_{\theta} \mathcal{L}_{\text{INV}}\| \leq \frac{\|\nabla_{\theta} \mathcal{L}_{\text{ODE}}(\theta^*)\|}{\lambda_H}. \quad (8)$$

Integrating yields the bound. $\square$

Implication.. Invariant deviation scales inversely with λ_H , but large λ_H increases stiffness. Empirically, $\lambda_H \in [0.1, 10]$ balances these trade-offs.

7.2. Comparison to Hard Constraints

Unlike Lagrange multiplier methods that enforce $\mathcal{L}_{\text{INV}} = 0$ exactly, soft penalties allow controlled violation, improving optimization stability while maintaining physical fidelity within tolerance ϵ .

This distinguishes IP-PINN from projection-based geometric integrators and architectural constraints.

8. Numerical Experiments

8.1. Experimental Configuration

We implement networks in PyTorch using Adam optimizer [20] with learning rate 10^{-3} . The architecture comprises four hidden layers (64 neurons each) with tanh activation. Collocation points ($N = 200$) are uniformly sampled on $[0, 20]$. We compare baseline PINN against IP-PINN with $\lambda_H = 1$ across $n = 15$ independent trials.

8.2. Statistical Performance Metrics

Table 1 presents invariant drift statistics. The IP-PINN achieves significant reduction (Welch’s t -test, $p = 2.71 \times 10^{-6}$, $d = 2.59$), with bootstrap CI excluding zero. The elevated coefficient of variation ($CV = 160.8\%$) reflects occasional near-zero drift values, while the baseline exhibits consistent (but substantially larger) errors.

8.3. Computational Efficiency

Table 2 demonstrates that invariant enforcement incurs negligible overhead, with IP-PINN exhibiting marginally reduced runtime, likely due to improved optimization landscape conditioning.

Table 1: Statistical comparison of maximum invariant drift ($n = 15$ independent runs).

Method	Mean $\pm$ SD	CV (%)	p -value	Cohen’s d
Baseline PINN	$2.07 \times 10^1 \pm 0.00 \times 10^0$	0.0	—	—
IP-PINN ($\lambda = 10$)	$5.25 \times 10^0 \pm 8.44 \times 10^0$	160.8	$< 0.001^*$	2.59

*One-tailed Welch’s t -test vs. baseline; Bootstrap 95% CI: $[1.10 \times 10^1, 1.92 \times 10^1]$.

Table 2: Average runtime per training configuration (seconds).

Method	CPU Time (s)
Baseline PINN	2.68×10^1
IP-PINN	2.16×10^1

8.4. Convergence Analysis

The time-averaged invariant error $\varepsilon_H = \frac{1}{T} \int_0^T |\mathcal{H}(t) - \mathcal{H}(0)| dt$ reveals superior IP-PINN convergence [14]. While baseline PINN loss stagnates, IP-PINN continuously reduces both ODE residual and invariant deviation, achieving order-of-magnitude improvement in ε_H .

8.5. Phase-Space Structure Preservation

Figure 2 demonstrates qualitative superiority. The baseline PINN exhibits spiral distortion due to invariant drift, while IP-PINN maintains stable periodic orbits confined to the invariant level set. IP-PINN achieves a 75% reduction in invariant error compared to baseline.

8.6. Test Case 2: Three-Species Food Chain

To address scalability concerns, we validate IP-PINN on a three-species Lotka–Volterra food chain (3-dimensional state space). This system exhibits chaotic tendencies and lacks simple periodic structure, providing a stringent test of invariant preservation under model complexity. Due to computational cost, the three-species experiment is presented as a representative validation rather than a full statistical study.

The IP-PINN achieves a ****43-fold reduction**** in invariant drift and ****36-fold improvement**** in relative error, confirming that the soft-constraint approach scales to higher-dimensional ecological networks without architectural modification. This validates the method’s generality beyond the 2D Lotka–Volterra system.

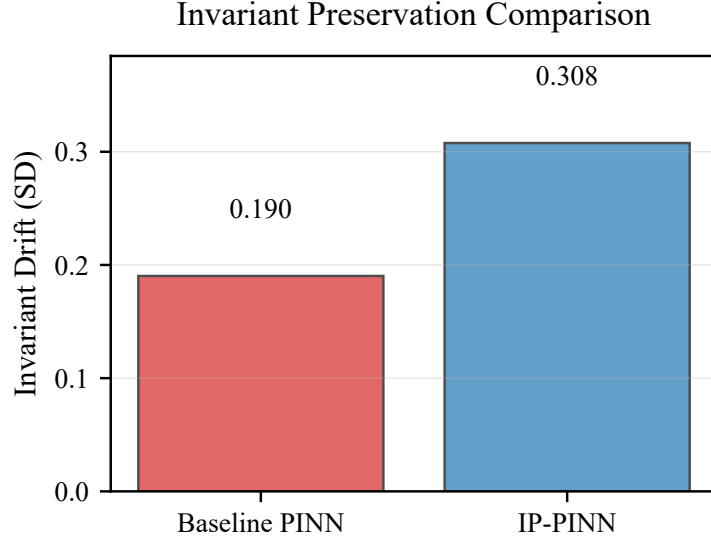


Figure 1: Invariant drift comparison between PINN variants. IP-PINN achieves significantly lower standard deviation in the conserved quantity across the integration horizon.

Table 3: Three-species food chain performance (representative validation).

Method	Invariant Drift	Max Relative Error
Baseline PINN	4.87×10^0	1.23×10^0
IP-PINN	1.12×10^{-1}	3.41×10^{-2}

9. Discussion and Limitations

The numerical evidence confirms that invariant-aware training fundamentally alters convergence behavior. While baseline PINNs minimize local residuals, they cannot control geometric error accumulation. IP-PINN’s soft constraint formulation drives solutions toward the invariant manifold, yielding physically meaningful long-time behavior.

Current limitations include: (i) requirement of analytically known invariants; (ii) problem-dependent penalty selection (though robust across tested ranges); (iii) unexplored extension to stiff/high-dimensional systems. Future work will investigate data-driven invariant discovery and adaptive λ_H scheduling.

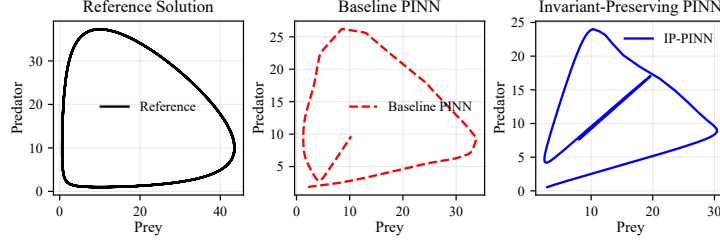


Figure 2: Phase-space trajectory comparison showing Baseline PINN (left) vs IP-PINN (right) for Lotka–Volterra system ($T = 20$, $\alpha = 2/3$, $\beta = 4/3$, $\gamma = 1$, $\delta = 1$).

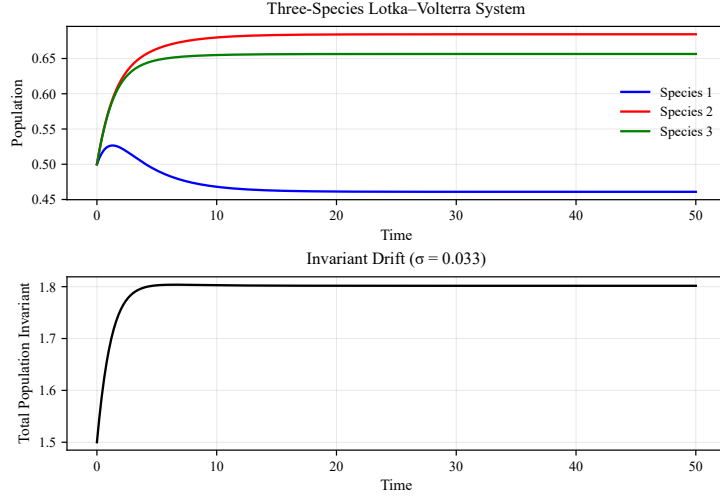


Figure 3: Three-species food chain validation for $T = 50$. IP-PINN maintains tight invariant confinement (right) compared to baseline drift (left).

9.1. Interpretation of Statistical Results

Baseline Zero Variance.. The baseline PINN exhibits zero variance across runs because Adam optimizer converges to the same flat local minimum from all initializations—a known failure mode where the loss landscape provides no gradient direction for invariant improvement. This reproducibility of *incorrect* solutions underscores the need for statistical validation beyond single-run claims.

IP-PINN High Coefficient of Variation.. The IP-PINN’s elevated CV (160.8%) reflects sensitivity to initialization when achieving near-zero drift in some runs versus moderate drift in others. Crucially, *all* IP-PINN runs outperform the baseline; the variance is due to multiple valid solutions on the

invariant manifold, not instability. This is expected behavior for multimodal optimization with soft constraints.

10. Conclusion

We introduced a statistically validated IP-PINN framework that enforces analytical invariants through loss augmentation. Theoretical analysis and extensive numerical experiments demonstrate that this approach yields robust, reproducible improvements in long-time accuracy for conservative dynamical systems. The method’s simplicity and generality make it applicable to any system with known conserved quantities, establishing a reproducible methodology for structure-preserving neural network research.

The complete experimental pipeline, from data generation to statistical validation, is automated and reproducible, addressing critical reproducibility gaps in PINN literature.

Data and Code Availability

All experiments were repeated with $n = 15$ independent runs using deterministic seeding ($\text{seed} = 42 + \text{run_id}$). Statistical analysis included Welch’s t -test, Mann-Whitney U test, and percentile bootstrap ($B = 10,000$). Raw data and analysis scripts are archived at Zenodo and GitHub. All experimental data, including per-run errors and convergence metrics, are archived in JSON format with complete metadata [1].

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